A MAP FROM A LINEARLY ADEQUATE FUNCTOR INTO AN ADEQUATE MODULE-VALUED FUNCTOR LIFTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file adequate.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The weight decomposition of module-valued functors over a graded algebra. Same chapter, Lemma lemma-flat-functor-split, invoked at proof lines 631-633. Gives L(B)^{(1)} and phi(xi) in M tensor_A B_1.
- The flat base change property for the functors occurring in the diagram. Same chapter, Lemma lemma-adequate-flat, invoked at proof lines 631-633. Supplies the hypothesis of the weight decomposition.

Source reference. Original lemma 06UW: sources/stacks-adequate/pinned-original-source.tex, source lines 442–442. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 06UZ: sources/stacks-adequate/pinned-original-source.tex, source lines 533–533. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let A be a ring. Given a solid diagram*

$$\begin{array}{ccccccc} 0 & \longrightarrow & L & \longrightarrow & \underline{A^{\oplus n}} & \longrightarrow & \underline{A^{\oplus m}} \\ & & \downarrow{\scriptstyle \varphi} & \swarrow & & & \\ & & \underline{M} & & & & \end{array}$$

of module-valued functors on Alg_A with exact row there exists a dotted arrow making the diagram commute.

Proof. Suppose that the map $A^{\oplus n} \to A^{\oplus m}$ is given by the $m \times n$-matrix (a_{ij}). Consider the ring $B = A[x_1, \ldots, x_n]/(\sum a_{ij} x_j)$. The element $(x_1, \ldots, x_n) \in \underline{A^{\oplus n}}(B)$ maps to zero in $\underline{A^{\oplus m}}(B)$ hence is the image of a unique element $\xi \in L(B)$. Note that

Date: stacks-project:adequate.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

ξ has the following universal property: for any A-algebra C and any $\xi' \in L(C)$ there exists an A-algebra map $B \to C$ such that ξ maps to ξ' via the map $L(B) \to L(C)$.

Note that B is a graded A-algebra, hence we can use Lemmas 06UZ and 06UW to decompose the values of our functors on B into graded pieces. Note that $\xi \in L(B)^{(1)}$ as $(x_1, \ldots, x_n)$ is an element of degree one in $\underline{A^{\oplus n}}(B)$. Hence we see that $\varphi(\xi) \in \underline{M}(B)^{(1)} = M \otimes_A B_1$. Since B_1 is generated by $x_1, \ldots, x_n$ as an A-module we can write $\varphi(\xi) = \sum m_i \otimes x_i$. Consider the map $A^{\oplus n} \to M$ which maps the ith basis vector to m_i. By construction the associated map $\underline{A^{\oplus n}} \to \underline{M}$ maps the element ξ to $\varphi(\xi)$. It follows from the universal property mentioned above that the diagram commutes. $\square$